

Orlando Alexander

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EDUCATION

BSc Data Science, *University of Bristol* (2024 - 2027)

- First-Class in Year 1: Top marks in Algorithms & Programming (87%) and Coding & Data Analysis (83%).
- Authored renewable energy paper (85%) using Python to analyse 28-year global dataset, revealing inverse GDP-fossil fuel consumption correlation and 8-fold renewable energy growth in high-income countries (2000-2015).
- Proficient in Python, JavaScript, React, SQL, R, C, C++, TensorFlow, Pandas, NumPy, Flask, AWS, GCP, and GitHub.

A-Levels, *Berkhamsted Sixth Form* (2020 - 2022)

- A*, A*, A*, A in Physics, Computer Science, Mathematics, and Further Mathematics.
- A* (100%) in EPQ: Designed a flow-optimised hydroelectric energy generator to maximise wastewater head.
- Head Boy: Founded Academic Gala; delivered talks to 1,000+ audience members; campaigned on food waste.

Data Science Bootcamp, *Le Wagon* (Oct - Dec 2022)

- Intensive 10-week course on Machine Learning, Deep Learning, Data Engineering, and Data Analysis.
- Capstone Project: Trained 9 CNN classifiers on 40,000 fashion images with TensorFlow (70–95% accuracy); integrated with ASOS API and KNN similarity model via Scikit-Learn to recommend personalised outfits.

PROFESSIONAL EXPERIENCE

Data Science & Software Lead, *180 Degrees Consulting, Bristol* (Sep 2024 - Present)

- Lead 40 data science & software consultants delivering pro-bono projects for 7 global social impact clients.
- Secured industry partnerships with Le Wagon and Accenture to deliver mentorship and technical training.
- **Data Pipeline for Disease Analysis**, *EcoSwell* (Jun - Sep 2025)
 - Implemented an end-to-end data pipeline consolidating 66,000 entries from 5 years of weather, dengue, census, and expenditure data using Pandas, NumPy, and Selenium, replacing weeks of manual downloading.
 - Used Monte Carlo analysis to forecast dengue outbreaks in Peru, enabling data-driven public health decisions.
- **Resource Management Platform**, *everyFAMILY* (Feb - Apr 2025)
 - Deployed modern React data management web app with Flask REST API backend (Docker on GCP Cloud Run).
 - Reduced onboarding time by 30%; delivered scalable platform rated 5/5 by client for 'exceeding expectations'.
- **Food Aid Analytics Dashboard**, *Cirencester Foodbank* (Sep - Dec 2024)
 - Developed analytics dashboard to process and visualise 5,000+ food aid records across 100 data fields, implementing Streamlit frontend, REST API backend, and Python data stack (Pandas, NumPy, Matplotlib).
 - Delivered 3 years of insights on regional demand and seasonal trends to enable targeted aid allocation.

Full-Stack Software Engineer, *Educatch, London* (May 2024 - Present)

- Designed and deployed production-grade React platform with MySQL backend, automating scheduling, invoicing, and attendance analytics for 50+ students.
- Reduced administrative workload by 10 hours/week; became company's most-used tool, supporting expansion.

Junior Software Engineer, *WISENSE, Berlin* (Feb 2023 - Feb 2024)

- Built ML studio with drag-and-drop interface to create video analysis workflows in <5 mins using React and Axios.
- Deployed for major clients including BMW, ABB, and SIG for real-time manufacturing anomaly detection.

Junior AI Engineer, *Connect AI* (Jan 2023 - Jan 2024)

- Developed GPT-4 LLM-powered customer support assistant with React frontend for startups including Digital Republic, handling 5,000 messages/month with 24/7 multilingual support.
- Implemented RAG (Retrieval-Augmented Generation) workflows with real-time feedback loops, reducing first-level ticket volume by 20% and improving resolution time by 95%.

Renewable Energy Intern, *EcoSwell, Peru* (Aug - Sep 2022)

- Developed integrated monitoring system with Python on Raspberry Pi, capturing 10,000+ data points weekly across 8 key data types.
- Designed offline logging system for automated, reliable reporting on UN SDG goals in remote conditions.

Computer Vision Engineer Intern, *Inovo Robotics, Guilford* (Aug 2021)

- Designed computer vision system for automated pastry decoration, 90% more efficient than manual methods.
- Implemented advanced algorithms including Canny edge detection, Hough transformations, and clustering for path optimisation using OpenCV and mlrose.

INTERESTS

Programming Tutor — Tutored 300+ students in Python, Web Development and ML at PMT, codetoday and Code Kids.

Drama — Performed at Edinburgh Fringe 2024; produced and acted in 15 Bristol DramSoc productions.